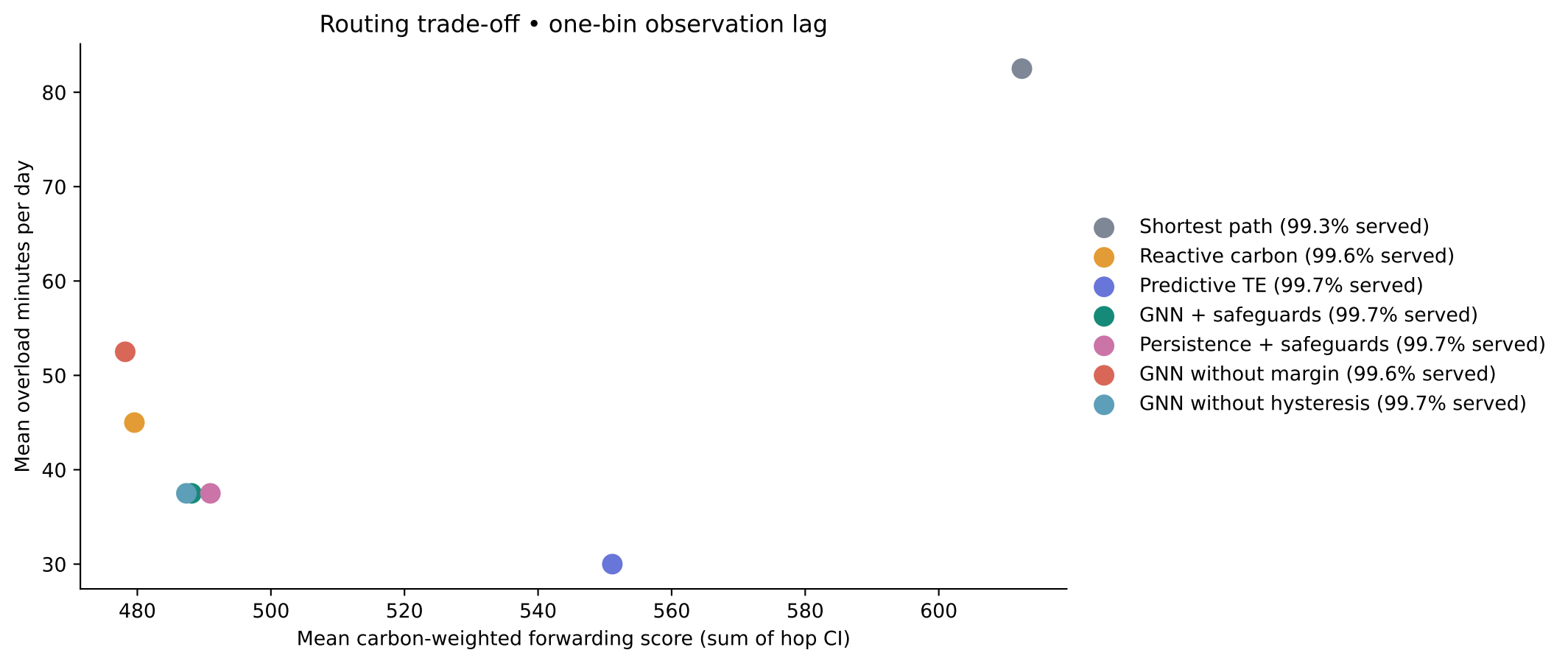
Routing trade-off • one-bin observation lag
Mean overload minutes per day
80
70
60
50
40
30
480
500
520
540
560
580
600
Mean carbon-weighted forwarding score (sum of hop CI)
Shortest path (99.3% served)
Reactive carbon (99.6% served)
Predictive TE (99.7% served)
GNN + safeguards (99.7% served)
Persistence + safeguards (99.7% served)
GNN without margin (99.6% served)
GNN without hysteresis (99.7% served)